

DBMS PROJECT

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Section: C

Project Title: Student Achievement Record System

Problem Statement No.: 29

1. Problem Statement

Students participate in various competitions and receive awards at different levels, but their achievement records may be scattered or maintained manually, making them difficult to manage and search.

The **Student Achievement Record System** provides a structured way to store, view, search, and update achievement details such as competition, award, year, and level. It helps maintain these records efficiently and makes **finding achievement information easier**.

2. Objectives

To develop a simple database-driven web application that stores and manages student achievement records using PHP and MySQL. The system allows users to efficiently add, view, search, and update achievement information.

Features

- Add student achievement records.
- View all saved achievement records.
- Search records by competition, award, year, or level.
- Update existing achievement records.
- Store records securely in a MySQL database.
- Use prepared statements for secure database operations.

3. Technology- HTML, CSS, PHP, MySQL, and InfinityFree.

4. Database Design

Database Name: student_achievement_db

Table Name: achievements

Column Name	Data Type	Constraints
achievement_id	INT	Primary Key, auto_increment
competition	VARCHAR(100)	NOT NULL
award	VARCHAR(100)	NOT NULL
year	INT	NOT NULL
level	VARCHAR(50)	NOT NULL

5. App Flow

HTML Form → PHP Server-side Program → MySQL Database → SELECT Result → Browser Output

Working Flow

- User enters achievement details through the HTML form.
- PHP receives and validates the form data.
- PHP stores the data in the MySQL database.
- PHP uses SELECT queries to retrieve records.
- The records are displayed in the browser.

6. SQL Operations

- **INSERT** – To add new achievement records.
- **SELECT** – To view and search achievement records.
- **UPDATE** – To edit existing achievement records.

Security

User input is handled using **prepared statements** to help prevent SQL injection.

7. Prepared Statements

Prepared statements are used to securely handle user input in SQL queries. They use placeholders (?) and parameter binding instead of directly adding user input to SQL queries. This helps prevent SQL injection.

8. Hosted Application

Hosting Platform: InfinityFree

Hosted URL: <https://dbms.fwh.is>

The Student Achievement Record System is hosted on InfinityFree and can be accessed through the above URL.

9. Test Cases / Testing

Test Case	Expected Result	Status
Add achievement record	Record saved successfully	Pass
View records	Saved records displayed	Pass
Search achievement	Matching records displayed	Pass
Update achievement	Record updated successfully	Pass
Security test using O'Reilly	Input handled safely	Pass

Home/Data Entry Form

Student Achievement Record System

Add and view student achievement records.

Add Achievement

View Records

Search Records

Achievement Entry Form

Competition:

Award:

Year:

Level:

Add Achievement

Activate Windows

Go to Settings to activate Windows.

Student Achievement Record System

Add and view student achievement records.

Add Achievement

View Records

Search Records

Achievement Entry Form

Competition:

Datathon

Award:

Second Prize

Year:

2026

Level:

College

Add Achievement

Activate Windows

Go to Settings to activate Windows.

Achievement Saved

Student Achievement Record System

Achievement Saved Successfully

Your achievement record has been added to the database.

[Add Another Achievement](#)

[View Records](#)

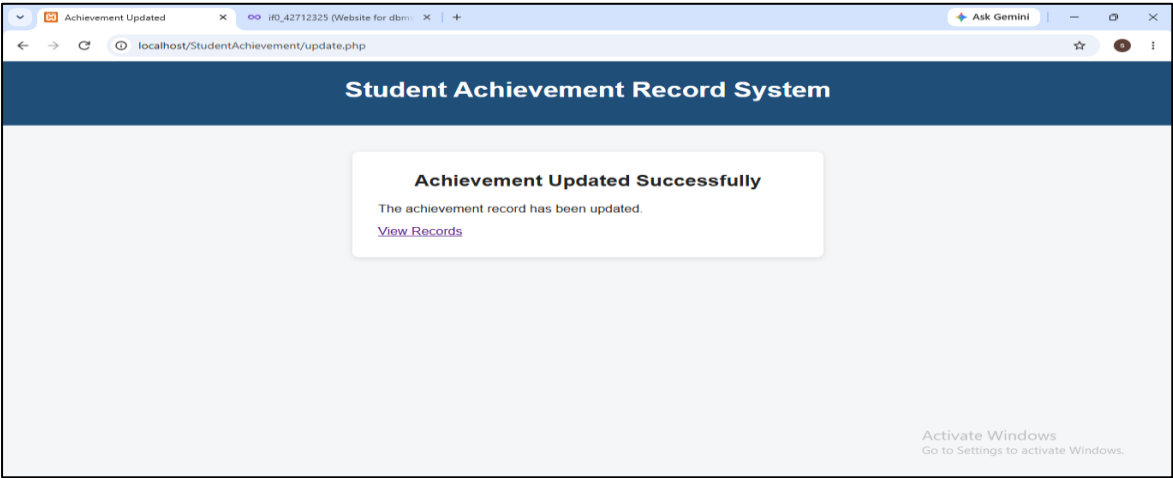
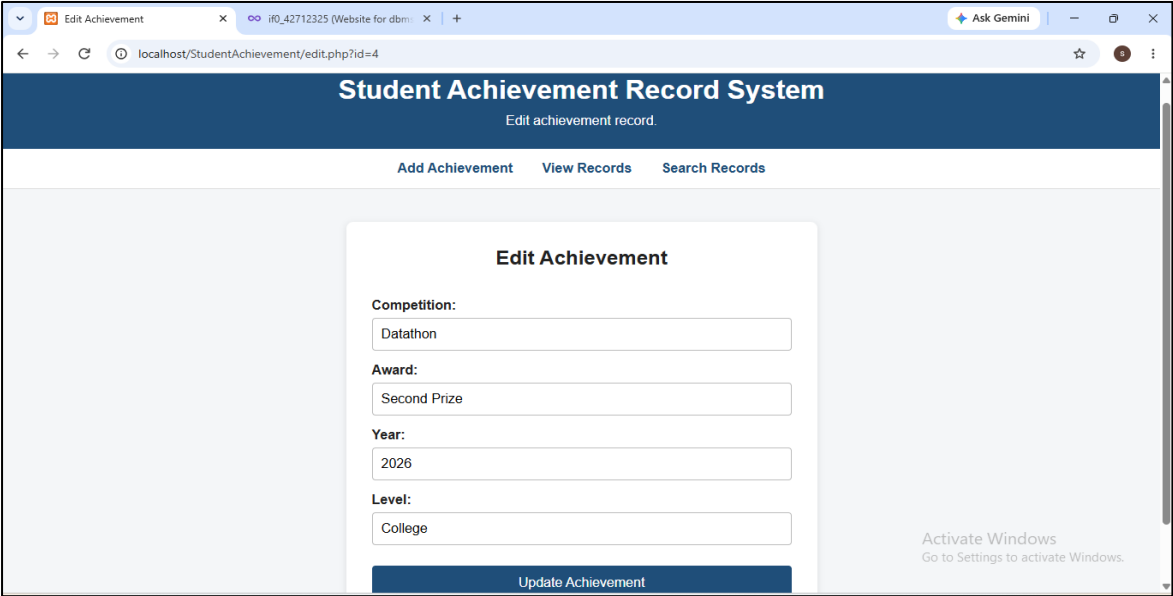
Activate Windows

Go to Settings to activate Windows.

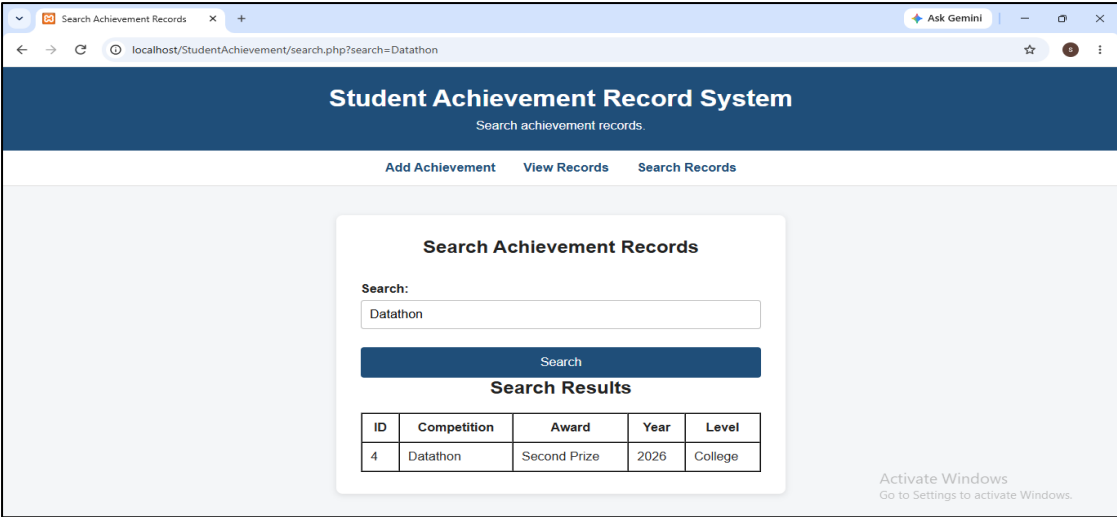
Record List



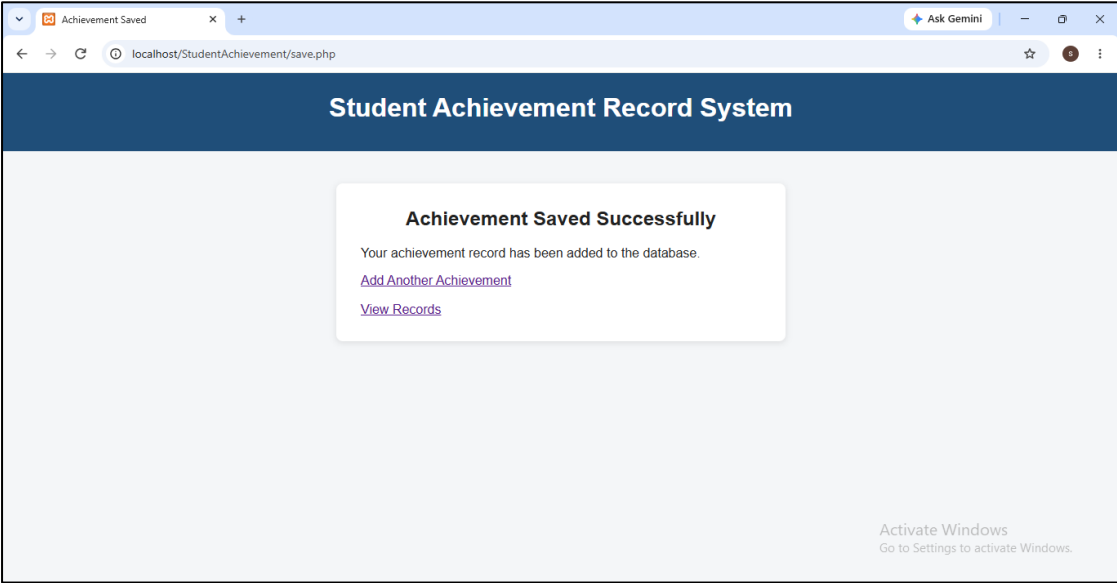
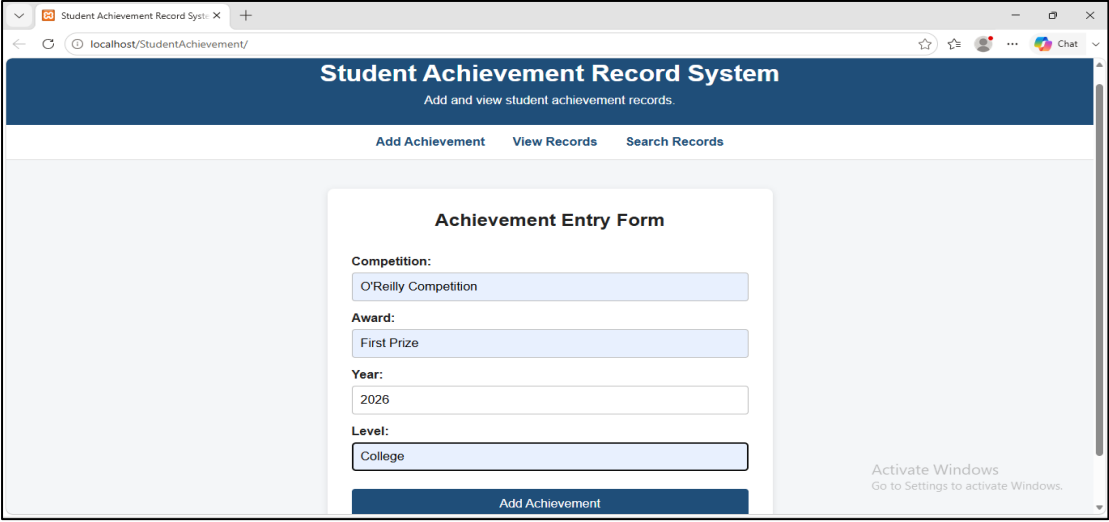
Edit/Update Record



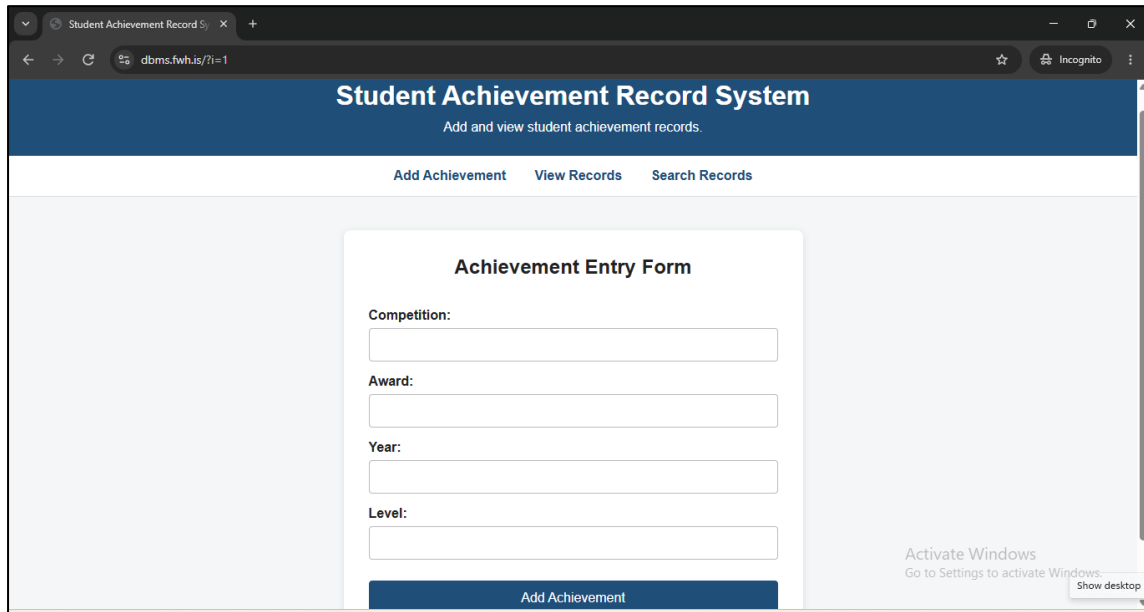
Search Records



Security Test Input/Success



Hosted URL



The screenshot shows a web browser window with the URL `dbms.fwh.is/7i=1`. The page title is "Student Achievement Record System" with the subtitle "Add and view student achievement records." Below the title is a navigation bar with three links: "Add Achievement", "View Records", and "Search Records". The main content area features a central "Achievement Entry Form" with four input fields labeled "Competition:", "Award:", "Year:", and "Level:". At the bottom of the form is a blue button labeled "Add Achievement". In the bottom right corner of the browser window, there is a Windows activation watermark that says "Activate Windows" and "Go to Settings to activate Windows", along with a "Show desktop" button.

10. Declaration

I confirm that the **Student Achievement Record System** has been developed by me as part of the DBMS Project using HTML, CSS, PHP, and MySQL.

